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Evaluating FlexTail: A Wearable Device for Spinal Posture Tracking

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Abstract:
Objectives: Accurate, continuous spinal posture tracking is critical for understanding musculoskeletal disorders and developing personalized treatment. However, current solutions are costly, obtrusive, or insufficiently accurate in daily life. This study quantifies the accuracy of FlexTail, a flexible, wearable sensor that integrates printed strain-gauge electronics to capture three-dimensional (3D) spinal motion in real-time.
Methods: FlexTail's performance is benchmarked with 3D-printed validation templates: three fixed-diameter S-shaped flexion paths (100, 150 and 250 mm flexion diameter), a spine-shaped path, and a 90° torsion path. The measured deformation is compared to the known geometry to quantify the precision of the device. We analyze ten FlexTail devices and repeat each measure ten times.
Results: Across flexion templates, the mean positional error is 3.61 mm~~3.76 mm~~. The 250 mm flexion template yields the lowest error, 2.12 mm. The mean torsional error is 0.13°, and the spine-shaped template shows an angular deviation of 1.05°~~1.00°~~ with a positional error of 2.95 mm~~3.27 mm~~. The intra-device deviations stay below 0.60 mm~~0.58 mm~~ and 0.051° for flexion and torsion, respectively.
Conclusions: FlexTail delivers reliable flexion and torsion tracking with accuracy equal to clinical systems. Its soft, integrated form factor supports long-term, unobtrusive use as a tool for biomechanical analysis and digital health applications.

Keywords: biomechanics; flexion measurement; health monitoring; printed electronics; sensor validation; wearable sensor

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Introduction

To achieve reliable and practical posture tracking remains a long-standing challenge in the health sector [1]. It plays a key role in the monitoring of surgical outcomes, the optimization of movement and the understanding of musculoskeletal disorders, but current methods lack standardization and reliability, limiting their applicability in real-world settings [2].

Although some studies suggest that spinal misalignments or specific postural behaviors contribute to back pain [3], others do not find a significant correlation [4]. Systematic reviews report conflicting findings, partly due to limitations in current posture tracking methods, which often rely on static assessments, self-reports, or laboratory-based observations [5, 6]. These approaches fail to capture the dynamic nature of posture in daily activities, making it unclear whether posture is a causal factor in back pain or merely correlated with other conditions. Recent studies suggest that static spinal assessment alone is insufficient as it does not account for muscle stiffness and spinal alignment, which cause back pain [7].

Several technologies attempt to address these limitations. Optical motion capture systems, such as Vicon (Vicon Motion System LTD, Oxford, UK), provide sub-millimeter accuracy but are confined to controlled laboratory environments, restricting their use for long-term tracking [8]. Wearable inertial measurement units (IMUs) offer a more practical alternative but are obtrusive [9] and suffer from sensor drift and integration errors, which significantly reduce their accuracy over time, particularly when estimating 3D positions from acceleration data [10]. Hybrid approaches, such as the combination of smartphone cameras and IMUs as shown by Wang et al., attempt to improve tracking by fusing multiple sensor inputs [11]. Although such approaches are promising, Wang et al. show that markerless systems are not yet able to accurately capture the spine's shape.

Alternative solutions include mechanical spinal measurement devices, such as Idiag M360 (formerly SpinalMouse, Idiag AG, Fehraltorf, Switzerland), which estimates spinal curvature based on spinous pro-

cess distances. However, such devices require trained operators, are limited to static assessments, and cannot track dynamic postural changes [12]. Raster stereography systems, such as Formetric III-4D (DIERS International GmbH, Schlangenbad, Germany), also provide non-invasive spinal shape analysis, but are limited to clinical settings [13].

Consumer electronics emerge as a potential solution for unobtrusive health tracking, although most devices focus on adjacent metrics such as step count or activity intensity rather than direct posture monitoring. The review by Haddas et al. [14] identifies only one consumer-grade wearable designed specifically for posture tracking. UprightGo (Upright Technologies LTD, Tel Aviv, Israel), for example, continuously tracks upper body posture, but relies on a single measurement point on the neck [15]. Similarly, Jordi Bruin's Posture Pal app leverages inertial sensors in modern headphones to monitor head position, but neither solutions provide comprehensive spinal shape tracking [16].

Other IMU-based wearable systems track posture with different sensor placements. Spineangel (Movement Metrics LTD, Hamilton, New Zealand) uses a single IMU mounted on the pelvis, while ViMove (dorsaVi LTD, Melbourne, Australia) uses two IMUs placed on the back [17, 18]. Another approach is strain-gauge-based sensing, such as BodyGuard (Sels Instruments NV, Vorselaar, Belgium), which measures lumbar spine curvature using flexible sensor technology [19]. While these systems allow for continuous monitoring, they provide only localized posture data rather than a full spinal shape analysis.

Despite the growing interest in posture-tracking technologies, no existing system enables continuous, unobtrusive, and real-time tracking of the complete three-dimensional (3D) shape of the spine. This gap highlights the need for a wearable, flexible device capable of capturing both flexion and torsion of the spine over time.

FlexTail (MinkTec GmbH, Braunschweig, Germany) is a wearable device that incorporates a flexible sensor strip aligned along the spine. Integrated into a tight-fitting shirt, it measures spinal flexion and torsion through 18 evenly distributed sensor elements. This enables real-time 3D posture tracking without relying on optical motion-capture systems or obtrusive inertial sensors or markers attached to the back. In contrast to inertial systems, which infer posture indirectly from segment orientation or acceleration at discrete attachment points, FlexTail reconstructs spinal posture directly from the measured geometric deformation of the sensor strip itself. Existing strain-based or IMU-based wearables typically provide posture information at one or two distinct locations, whereas the distributed sensing approach of FlexTail covers the full lumbar and most of the thoracic spine. The single-strip design further simplifies device placement and enables independent setup without external assistance.

This work evaluates the FlexTail device designed for continuous spinal posture tracking. We contribute:

- A general evaluation concept for posture sensors.
- Comprehensive assessment of the FlexTail's accuracy.
- Evaluation of its suitability for relevant use-cases.

Materials and Methods

FlexTail Device

The FlexTail device is composed of two parts (Figure 1): a small housing with battery and the sensing strip containing the $S = 18$ strain-gauge-based sensor elements. The strip has a length of 457 mm, a width of 26 mm, and a depth of 0.8 mm. To accommodate different spine lengths, the strip can be shortened by trimming off sensor elements from the end. This is possible due to the printed electronics architecture, which addresses each sensor individually. In this study, we evaluate the full-length configuration to assess the device's maximum sensing capacity. The distance between the sensor elements is 25 mm. We define a right-handed coordinate system (x, y, z) with its origin located inside the electronics housing. The z-axis follows the longitudinal direction of the sensing strip, the x-axis points laterally outward, and the y-axis points orthogonally downward, forming the reference frame for all flexion and torsion measurements.

The FlexTail connects to a smartphone via Bluetooth Low Energy, allowing real-time data recording through an app. The app enables calibration of the device, grouping and labeling of the measurements, and exporting the data into a comma-separated value (CSV) file. The maximum sampling rate of the system is 100 Hz, though lower rates (e.g., 5 Hz) significantly extend operational time.

To assess unit-to-unit variations, we choose $D = 10$ devices randomly from the production. We validate the devices' accuracy and reproducibility in three separate experiments: (i) large flexion, (ii) flexion of a typical human spine, and (iii) large torsions.

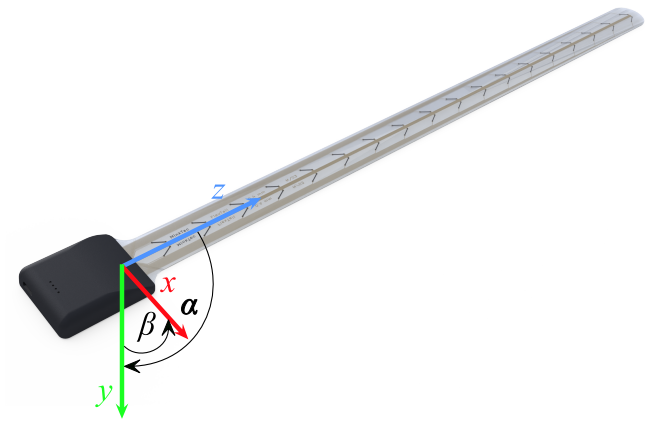

Fig. 1: The FlexTail device consists of a compact electronics housing and a flexible sensing strip. The coordinate system (x, y, z) originates inside the housing, defining directions for flexion (angle α) and torsion (angle β).

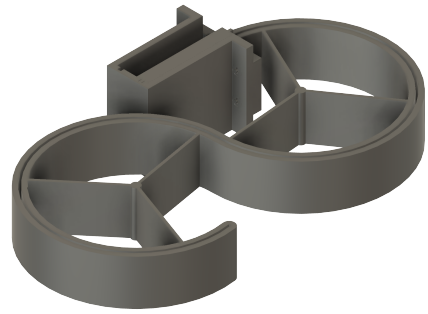

Fig. 2: 3D model of the $100\,\mathrm{mm}$ diameter flexion template.

Fabrication of Templates

For all experiments, we create templates using a 3D printer (Ender 3 V2, Creality, Shenzhen, China) and Polylactide (PLA, Elegoo, Shenzhen, China) for the template's material due to its advantageous properties: its high stiffness minimizes deformation during device insertion and its low warping characteristics preserves the geometric accuracy. Furthermore, PLA prints easily, reducing manufacturing errors [20].

Design of Templates

We choose fused deposition modeling (FDM) over stereolithography (SLA) for its ability to maintain dimensional stability after hardening, preventing geometric distortions that can occur in SLA prints as a result of resin curing effects [21].

Function of Templates

Each template guides the sensing strip along a predefined, geometrically determined path and ensures an exact fit to prevent slippage or misalignment during the measurements. Therefore, it has a slit (500 mm length, 1.2 mm width, 26 mm and 27 mm height for flexion and torsion, respectively) ensuring simple handling and precise fitting despite printing tolerances.

In addition, we use a bracket to secure the FlexTail's housing at the front opening of each template, preventing misalignment but enabling flipping to measure in both directions.

Accuracy of Templates

We determine the printing accuracy of the FDM printer by measuring the smallest template's actual diameter with a caliper and comparing it to the desired diameter, finding an error of $\delta_\mathrm{p} = 0.15\,\mathrm{mm}$. Given that this deviation is small relative to the expected sensor errors, we proceed without explicitly correcting it.

Shape of Templates

Fixed-Diameter Flexion Templates

We design three templates with distinct flexion diameters to evaluate the device's performance under varying curvatures. The shape turns at the center of the template: the flexion direction reverses, while the flexion diameter remains. This configuration bends the sensing strip into an S-shape without introducing torsion, applying curvature in the y–z plane and thereby defining flexion as rotation around the x-axis (angle α).

The diameters are 100 mm (Figure 2), 150 mm, and 250 mm. The 100 mm template applies the highest flexion specified by the device's manufacturer. The 150 mm and 250 mm templates serve as intermediate stages, bridging the gap between the highest flexion and the regular spine-shaped flexion.

Spine-shaped Flexion Template

To examine the typical use-case, i.e., monitoring the human spine in the sagittal plane, we design a template in the average healthy spine's shape in this plane.

According to Vrtovec, Pernuš & Likar [22] and Been & Kalichman [23], we follow the approaches of Janik et al. [24] and Harrison et al. [25] and geometrically define the curvature of the lumbar lordosis and the thoracic kyphosis using best-fitting ellipses (Figure 3). We prefer this parametric model over raw spine data because the closed-form ellipse lets us compare sensor readings to the template with straightforward, analytic calculations.

We select the T1–T12 thoracic kyphosis model reported by Harrison et al. to include the thoracolumbar junction at T12 and ensure geometric continuity with the lumbar lordosis model.

For lumbar lordosis and thoracic kyphosis, we set the semi-major axis to $a_1 = 154.3\,\text{mm}$ and $a_2 = 276.1\,\text{mm}$, respectively, and the semi-minor axis to $b_1 = 60\,\text{mm}$ and $b_2 = 187\,\text{mm}$, respectively. The quadrant portion of the ellipse is $\phi_1 = 81.2°$. We chose these parameters following the three-parameter model for the region T1–T12 of Harrison et al. with a quadrant portion of $\phi_2 = 72.2°$ [25]. We choose this model over the four-parameter formulation to enforce a shared anatomical reference at T12: fixing the horizontal position of the ellipse center constrains the semi-minor axes of the thoracic and lumbar ellipses to pass through the respective T12 endplates, establishing a common vertical reference axis and consistent geometric alignment across the thoracolumbar junction.

Both ellipses are the result of averaging the coordinates of the vertebral corner from radiographic images of 50 individuals for one ellipse and 80 for the other. These elliptical paths trace the posterior body margins from approximately S1 to T1, representing population-based reference geometries that capture the mean sagittal spinal curvature reported across multiple independent normative studies, rather than subject-specific spinal shapes [25, 24].

We set the semi-minor axis of the lumbar ellipse to align parallel to the inferior body endplate of T12, while the semi-minor axis of the thoracic ellipse aligns parallel with the posterior endplate of this vertebra. We set the height of T12 to $h_{\text{T12}} = 25\,\text{mm}$, which matches the distance between two sensor elements and meets the range reported by Panjabi et al. [26].

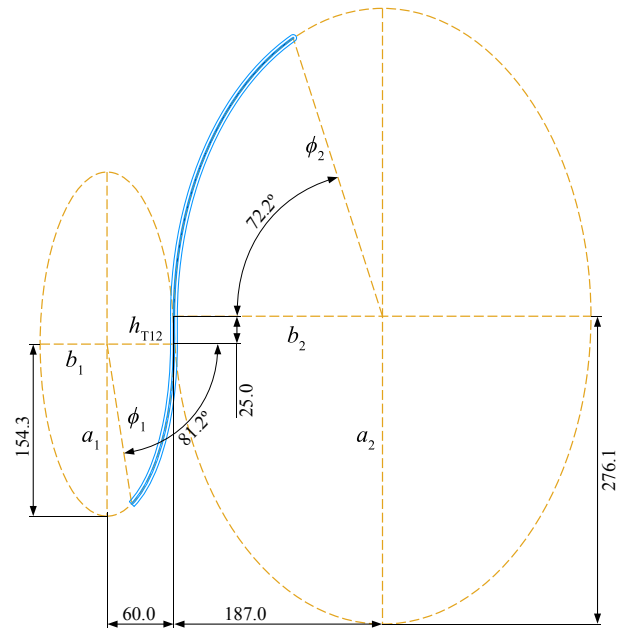

Fig. 3: Geometric construction of the spine-shaped template using best-fitting ellipses for the lumbar lordosis and thoracic kyphosis.

Torsion Template

We design a single template to evaluate torsional deformation, as flexion is the primary focus of this study (Figure 4). The template twists the sensing strip coaxially and uniformly around the z-axis by an angle of $\Theta = 90°$, ensuring that no flexion occurs and that the applied strain only results from axial rotation, unaffected by lateral or sagittal deviations.

Measurement Procedure

We design all flexion templates to insert the device from the side into the template. This ensures that each segment bends in a single direction without first flexing in the opposite direction, as when inserted along the template. For the torsion template, we insert the device from the bracket-side of the template.

All devices remain in the template for 3 s while we record the data continuously at 25 Hz, which is the maximum sampling rate the software supports. This duration balances between capturing sufficient data for reliable analysis and minimizing the measurement time.

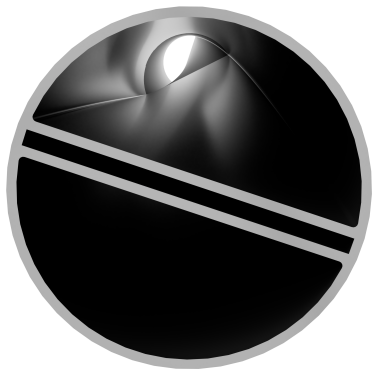

Fig. 4: 3D model of the torsion template.

We measure each device $d = 1, 2, ..., D$ ten times in one direction, flip it, and measure another ten times in the other direction, yielding a total of 20 repetitions per device and template. With continuous recording at 25 Hz over 3 s per repetition, this results in $T = 1500$ measurements per device and template. ~~For the flexion and torsion templates, $D = 10$, but due to device availability, $D = 4$ for the spine-shaped template.~~

Following the manufacturer's specifications, we calibrate the devices before each measurement sequence by straightening them on a flat surface and zeroing the readings.

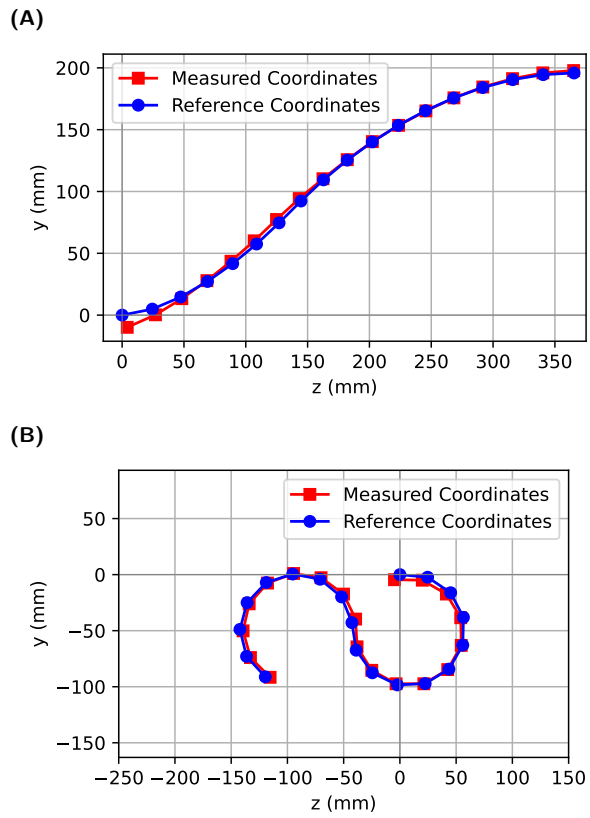

Fig. 5: Measured (red squares) and reference (blue circles) coordinates for a single device in (A) the spine-shaped template ($\text{MAE}_d = 2.94\,\text{mm}$) and (B) the 100 mm flexion template ($\text{MAE}_d = 4.95\,\text{mm}$).

Data Processing and Error Analysis

For each sensor element $s = 1, 2, \ldots, S$, $S = 18$, we collect two angles: α (flexion), representing rotation around the y-axis, and β (torsion), representing rotation around the z-axis.

Flexion Error

We evaluate the device's flexion accuracy by computing the 3D coordinates of all s from the measured angles using the Denavit-Hartenberg transform. We then compare the coordinates to the corresponding reference points along the template's geometric path (Figure 5). Before calculating the error, we center measured and reference paths and align them along their principal axes.

We calculate the positional error $E_{\text{pos},d,s,t}$ as the Euclidean distance between each measured coordinate $\vec{x}_{\text{m},d,s,t}$ and its corresponding reference coordinate $\vec{x}_{\text{r},s}$:

$$E_{\text{pos},d,s,t} = \|\vec{x}_{\text{m},d,s,t} - \vec{x}_{\text{r},s}\|_2. \tag{1}$$

We calculate the angular error $E_{\text{ang},d,s,t}$ by subtracting the measured alpha angle $\alpha_{\text{m},d,s,t}$ from the reference angle $\alpha_{\text{r},s}$:

$$E_{\text{ang},d,s,t} = \alpha_{\text{m},d,s,t} - \alpha_{\text{r},s}. \tag{2}$$

Torsional Error

The torsion template has a gradual rotation along the device's longitudinal axis, resulting in a total angle of $\Theta = 90°$. We determine the reference rotation angle per sensor element as:

$$\beta_{\text{r}} = \frac{\theta}{S} = \frac{90°}{18} = 5°. \tag{3}$$

We calculate the torsional error $E_{\text{tor},d,s,t}$ by subtracting the measured beta angles $\beta_{\text{m},d,s,t}$ with the reference values β_{r}:

$$E_{\mathrm{tor},d,s,t} = \beta_{\mathrm{m},d,s,t} - \beta_{\mathrm{r}}. \tag{4}$$

Error Metrics

To assess accuracy and reproducibility, we compute the mean absolute error (MAE), root mean square error (RMSE), and standard deviation of the errors (σ) for both positional and angular errors.

The MAE for a single device d is:

$$\mathrm{MAE}_d = \frac{1}{T \cdot S} \sum_{t=1}^{T} \sum_{s=1}^{S} |E_{d,s,t}|. \tag{5}$$

where $E_{d,s,t}$ represents the individual measurement error (positional or angular) for each sensor element s of device d at measurement instance t.

To extend the MAE calculation across multiple tested devices, we compute:

$$\mathrm{MAE}_D = \frac{1}{D \cdot T \cdot S} \sum_{d=1}^{D} \sum_{t=1}^{T} \sum_{s=1}^{S} |E_{d,s,t}|. \tag{6}$$

Similarly, the RMSE for a single device and across multiple devices is:

$$\mathrm{RMSE}_d = \sqrt{\frac{1}{T \cdot S} \sum_{t=1}^{T} \sum_{s=1}^{S} (E_{d,s,t})^2}. \tag{7}$$

and

$$\mathrm{RMSE}_D = \sqrt{\frac{1}{D \cdot T \cdot S} \sum_{d=1}^{D} \sum_{t=1}^{T} \sum_{s=1}^{S} (E_{d,s,t})^2}. \tag{8}$$

respectively. Accordingly, the standard deviation of the errors for a single device and across multiple devices is:

$$\sigma_d = \sqrt{\frac{1}{T \cdot S} \sum_{t=1}^{T} \sum_{s=1}^{S} (E_{d,s,t} - \mathrm{MAE}_d)^2}. \tag{9}$$

and

$$\sigma_D = \sqrt{\frac{1}{D \cdot T \cdot S} \sum_{d=1}^{D} \sum_{t=1}^{T} \sum_{s=1}^{S} (E_{d,s,t} - \mathrm{MAE}_D)^2}. \tag{10}$$

Using equations (6), (8), and (10), we compute the template metrics ($\mathrm{MAE}_{\mathrm{tpl}}$, $\mathrm{RMSE}_{\mathrm{tpl}}$, and σ_{tpl}) by setting $D = 10$ for the flexion~~fixed-diameter~~ and torsion templates~~, and $D = 4$ for the spine-shaped template~~, while incorporating all measurements recorded within each respective template.

These metrics characterize FlexTail's performance in capturing the flexion or torsion characteristics of each template. Finally, the total metrics ($\mathrm{MAE}_{\mathrm{tot}}$, $\mathrm{RMSE}_{\mathrm{tot}}$, and σ_{tot}) aggregate all measurements across all devices and templates.

Intra-Device Deviation

To assess the variation between repeated measurements of the same device, we introduce the mean intra-device deviation (MID). The MID quantifies the differences in sensor readings between consecutive measurements, providing a measure of how consistently a device reproduces its own readings under identical conditions.

For positional consistency, we determine the Euclidean distance between the measured 3D coordinates of corresponding sensor elements in consecutive measurements:

$$\mathrm{MID}_{\mathrm{flex,pos},d} = \frac{1}{(T-1) \cdot S} \sum_{t=1}^{T-1} \sum_{s=1}^{S} \|\vec{x}_{\mathrm{m},d,s,t+1} - \vec{x}_{\mathrm{m},d,s,t}\|_2 \tag{11}$$

For angular reproducibility, we compute MID as the absolute difference between consecutive flexion angles:

$$\mathrm{MID}_{\mathrm{flex,ang},d} = \frac{1}{(T-1) \cdot S} \sum_{t=1}^{T-1} \sum_{s=1}^{S} |\alpha_{\mathrm{m},d,s,t+1} - \alpha_{\mathrm{m},d,s,t}| \tag{12}$$

and torsion angles:

$$\mathrm{MID}_{\mathrm{tor,ang},d} = \frac{1}{(T-1) \cdot S} \sum_{t=1}^{T-1} \sum_{s=1}^{S} |\beta_{\mathrm{m},d,s,t+1} - \beta_{\mathrm{m},d,s,t}|. \tag{13}$$

To generalize MID across multiple devices, we compute the mean of the per-device intra-device deviations:

$$\mathrm{MID}_D = \frac{1}{D} \sum_{d=1}^{D} \mathrm{MID}_d \tag{14}$$

Using this equation, we compute the template-specific and total intra-device deviation metrics ($\mathrm{MID}_{\mathrm{tpl}}$ and $\mathrm{MID}_{\mathrm{tot}}$) by incorporating all measure-

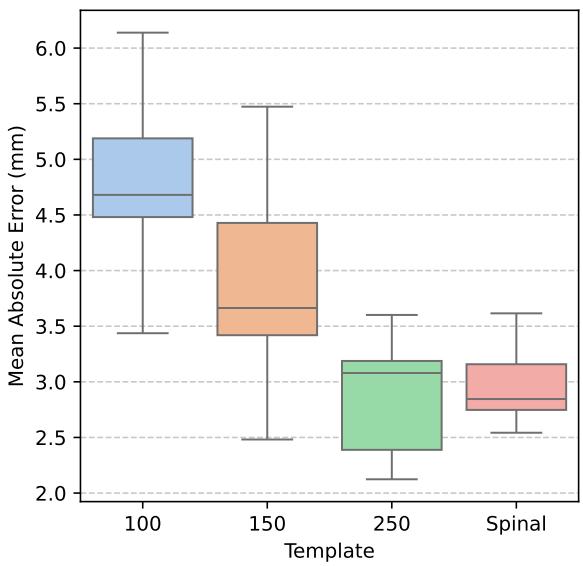

Fig. 6: Box plots of the positional MAE_d for the different flexion templates. Figure updated for new results.

ments from devices within each respective template and across all devices and templates, respectively.

Results

Across all flexion templates and measurements, the MAE_{tot} is 3.612 mm~~3.759 mm~~ for positional and 1.441° ~~1.506 °~~ for angular errors. The $RMSE_{tot}$ is 4.702 mm ~~4.879 mm~~ and 2.135°~~2.236 °~~, respectively, with a σ_{tot} of 3.011 mm~~3.111 mm~~ and 2.118°~~2.219 °~~. The performance does not differ significantly between flexion directions.

The template with the smallest flexion diameter (100 mm) exhibits the highest MAE_{tpl} of 4.768 mm. Increasing the flexion diameter to 150 mm reduces MAE_{tpl} to 3.811 mm and for 250 mm to 2.883 mm. The spine-shaped template yields MAE_{tpl} = 2.952 mm ~~3.273 mm~~ (Table 1).

The box plots in section 6 illustrate the variability of MAE_d between devices for each template. The median of the individual MAE_d values is lowest for the spine-shaped template (2.845 mm)~~250 mm template (3.079 mm)~~, followed by the 250 mm template (3.079 mm)~~spine-shaped template (2.845 mm)~~, the 150 mm template (3.664 mm), and the 100 mm template (4.680 mm). The interquartile range (IQR) is smallest for the spinal template (0.411 mm~~0.224 mm~~), indicating minimal device-to-device variation.

For the torsion template, the MAE_{tpl} is 0.128°, with an $RMSE_{tpl}$ of 0.194° and σ_{tpl} of 0.146° (Table 2).

Tab. 1: Average and best positional error metrics for different templates.

Average positional error			
Template	MAE_{tpl}	$RMSE_{tpl}$	σ_{tpl}
100	4.768 mm	5.904 mm	3.482 mm
150	3.811 mm	5.002 mm	3.239 mm
250	2.883 mm	3.643 mm	2.228 mm
spinal	~~3.273 mm~~	~~4.391 mm~~	~~2.927 mm~~
	2.952 mm	3.864 mm	2.493 mm
Best positional error (individual devices)			
Template	MAE_d	$RMSE_d$	σ_d
100	3.437 mm	4.209 mm	2.431 mm
150	2.482 mm	3.188 mm	2.001 mm
250	2.124 mm	2.706 mm	1.676 mm
spinal	~~2.941 mm~~	~~3.909 mm~~	~~2.576 mm~~
	2.542 mm	3.161 mm	1.878 mm

Tab. 2: Average and best angular error metrics for different templates.

Average angular error			
Template	MAE_{tpl}	$RMSE_{tpl}$	σ_{tpl}
100	2.149°	3.155°	3.152°
150	1.409°	1.948°	1.926°
250	1.163°	1.602°	1.564°
spinal	~~1.003 °~~	~~1.312 °~~	~~1.247 °~~
	1.045°	1.386°	1.352°
torsion	0.128°	0.194°	0.146°
Best angular error (individual devices)			
Template	MAE_d	$RMSE_d$	σ_d
100	1.935°	3.000°	2.991°
150	1.193°	1.730°	1.725°
250	0.966°	1.435°	1.417°
spinal	~~0.911 °~~	~~1.169 °~~	~~1.108 °~~
	0.842°	1.173°	1.127°
torsion	0.059°	0.088°	0.065°

For the 100 mm template, the lowest positional MAE_d observed across all devices is 3.437 mm, with an $RMSE_d$ of 4.209 mm and a σ_d of 2.431 mm. The lowest positional error for the 250 mm template MAE_d = 2.124 mm, $RMSE_d$ = 2.706 mm, and σ_d = 1.676 mm. In the spine-shaped template, the best-performing device records a positional MAE_d of 2.542 mm~~2.941 mm~~, with

an RMSE_d of 3.161 mm~~3.909 mm~~ and a σ_d of 1.878 mm ~~2.576 mm~~ (Table 1).

For angular measurements, the lowest MAE_d of an individual device is 0.842°~~0.911 °~~ in the spine-shaped template, with an RMSE_d of 1.173°~~1.169 °~~ and a σ_d of 1.127°~~1.108 °~~. The lowest torsional MAE_d is 0.059°, with an RMSE_d of 0.088° and σ_d of 0.065° (Table 2).

$\mathrm{MID}_{\mathrm{tot}}$ is 0.599 mm~~0.581 mm~~ for positional measurements and 0.033° for angular measurements. The $\mathrm{MID}_{\mathrm{tpl}}$ for positional measurements is 0.332 mm for the 100 mm template, 0.561 mm for the 150 mm template, 0.672 mm for the 250 mm template, and 0.829 mm ~~1.049 mm~~ for the spine-shaped template.

For angular measurements, $\mathrm{MID}_{\mathrm{tpl}}$ is 0.027° for the 100 mm template, 0.029° for the 150 mm template, 0.034° for the 250 mm template, and 0.043°~~0.053 °~~ for the spine-shaped template. For torsional measurements, $\mathrm{MID}_{\mathrm{tpl}}$ is 0.051°.

Discussion

With an angular $\mathrm{MAE}_{\mathrm{tpl}}$ of about 1° in spinal-like shapes and below 2.2° for maximal flexion, the FlexTail outperforms devices such as Spineangel (5° – 6° MAE) [27, 17] and BodyGuard (3° MAE) [19], both of which align with commonly accepted clinical thresholds for lumbar motion assessment. For monitoring lumbar flexion, FlexTail performs similar to ViMove (1.82° RMSE) [18].

The Idiag M360, a clinical tool frequently applied in radiation-free spinal assessment [28], reports inter-rater differences of 2.1° [29] and a precision of 1° [28]. While inter-rater variability does not directly compare to FlexTail's accuracy metric, it provides insight into the expected variability in clinical usage. The angular $\mathrm{MAE}_{\mathrm{tpl}}$ of the FlexTail for the spine-shaped template aligns with the precision limit of the M360. As clinicians consider the M360 to be reliable for static spine assessments [30], FlexTail supports continuous posture tracking with similar precision.

The thresholds for clinically meaningful changes in spinal alignment, as reported in the literature, are as follows: For lumbar lordosis, minimal detectable changes (MDCs) range from 3.9° (radiographic Cobb angles) [31] to 5.9° (inclinometry) [32]. Thoracic kyphosis shows similar MDCs of 5° to 9° [33, 34]. In cervical spine studies, minimal clinically important differences for single-level angles are 4° to 6° [35], confirming the observed deviations lie well below clinically relevant thresholds. With a minimum of 1.045° (spinal template) and a maximum of 2.149° (100 mm template), our measurements of angular measurements fall below these values. However, there are no clinical benchmarks of spinal curvature assessment for spatial (millimeter-based) accuracy.

Its positional accuracy, however, cannot be compared to the aforementioned systems, as these devices are not designed to reconstruct spinal geometry but rather aim to assess specific metrics such as segmental inclination or lumbar flexion.

Interestingly, torsional accuracy surpasses flexion accuracy, suggesting that FlexTail's torsional sensing mechanism is more robust.

However, we assess accuracy using predefined geometric templates rather than human subjects. Since the device's average error is an order of magnitude larger than the printing deviation of $\delta_{\mathrm{p}} = 0.15$ mm, we neglect the impact of printing errors. Unlike studies that validate devices directly on humans against X-ray or motion capture systems, our approach lacks direct comparison with a gold standard such as Vicon (0.153 mm MAE, 0.154 mm RMSE) [36], limiting the generalizability of our results.

Further validation on humans must address inter-session and inter-user variability, which is introduced by slight differences in garment placement. We need to quantify this variability using a gold standard such as motion capture (e.g., Vicon) during predefined postures and range-of-motion tasks and address inter-session effects by comparing repeated calibration poses within and across sessions, while inter-user variability requires repeated donning on the same subject under controlled conditions.

Furthermore, we need to assess the stability of the device under long-term continuous use and various temperature conditions. This will be done in future evaluation studies.

The spine-shaped template represents a population-based reference geometry rather than an exact anatomical replica of an individual spine. In vivo deviations due to soft tissue, skin, and garment compliance are negligible relative to inter-individual spinal variability and the measurement error of this bench-top validation.

Furthermore, the fixed length of the sensing strip (457 mm) restricts full spinal coverage, usually capturing the lumbar and thoracic spine but not the cervical region. ~~As another limitation, we include only four devices for measurements with the spine-shaped template, potentially affecting statistical significance.~~

The trends observed in the $\mathrm{MAE}_{\mathrm{tpl}}$ for different templates further highlight the differences in the propagation of positional and angular error. Unlike the positional $\mathrm{MAE}_{\mathrm{tpl}}$, the angular $\mathrm{MAE}_{\mathrm{tpl}}$ is lower in the spinal template than in the 250 mm template. This difference arises from the way errors propagate in positional and angular measurements. While positional errors accumulate across sensor elements due to their sequential computation in 3D space, angular errors are independent for each sensor element by comparing measured and reference angles. This explains why the trends in positional and angular $\mathrm{MAE}_{\mathrm{tpl}}$ differ.

The lowest positional $\mathrm{MAE}_{\mathrm{tpl}}$ is in the 250 mm template but not in the spinal template, probably due to challenges in radially aligning the measured and reference 3D coordinates. As the flexion diameter increases and the path straightens, achieving precise radial alignment becomes more difficult, particularly at the endpoints of the path, leading to increased errors (Figure 5).

~~The reduced variance in the spinal template results from the smaller sample size ($D = 4$), potentially under-representing inter-device variability.~~

The increase in mean intra-device deviation with larger flexion diameters contrasts with the decreasing trend of error metrics. This may result from the mechanical interaction between the sensing strip and the template. Smaller flexion diameters create higher forces, ensuring a tighter fit and reducing alignment variability. In contrast, larger diameters may allow greater positional variation, increasing deviation between repeated measurements. The higher intra-device deviation in the torsion template suggests that rotational deformation causes more variability than flexion, despite achieving higher overall accuracy.

In summary, FlexTail demonstrates strong potential for real-world posture tracking, but further validation on human subjects remains essential to confirm its applicability in clinical and ergonomic settings. In particular, we need to apply the device in preoperative evaluations, postoperative monitoring, and treatment planning when combined with radiographic imaging and patient-reported outcome scores [14]. Other wearable systems, such as Spineangel and BodyGuard, report higher measurement errors but align with commonly accepted clinical thresholds for lumbar motion assessment [18].

Conclusion

- Accuracy and reliability of the FlexTail device meet clinical requirements.
- Compared to existing wearable solutions, FlexTail combines flexibility, portability, and accuracy, addressing key limitations of current methods.
- However, we validate under controlled conditions and further investigations on human subjects are necessary.

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References

[1] Paul Schober. Lendenwirbelsäule Und Kreuzschmerzen. Münchener Medizinische Woche. 1937;84:336-8.
[2] Du SH, Zhang YH, Yang QH, Wang YC, Fang Y, Wang XQ. Spinal Posture Assessment and Low Back Pain. EFORT Open Reviews. 2023 Sep;8(9):708-18.
[3] Irvin RE. Invited Response (The Postural Structural Model, with Boundary Conditions). Journal of Bodywork and Movement Therapies. 2011 Apr;15(2):144-8.
[4] Kripa S, Kaur H. Identifying Relations between Posture and Pain in Lower Back Pain Patients: A Narrative Review. Bulletin of Faculty of Physical Therapy. 2021 Dec;26(1):34.
[5] Swain CTV, Pan F, Owen PJ, Schmidt H, Belavy DL. No Consensus on Causality of Spine Postures or Physical Exposure and Low Back Pain: A Systematic Review of Systematic Reviews. Journal of Biomechanics. 2020 Mar;102:109312.
[6] Lederman E. The Fall of the Postural-Structural-Biomechanical Model in Manual and Physical Therapies: Exemplified by Lower Back Pain. Journal of Bodywork and Movement Therapies. 2011 Apr;15(2):131-8.
[7] Masaki M, Ikezoe T, Yanase K, Ji X, Umehara J, Aoyama J, et al. Association of Pain History and Current Pain With Sagittal Spinal Alignment and Muscle Stiffness and Muscle Mass of the Back Muscles in Middle-aged and Elderly

Women. Clinical Spine Surgery: A Spine Publication. 2019 Aug;32(7):E346-52.

[8] Ployon A, Lavaste F, Maurel N, Skalli W, Roland-Gosselin A, Dubousset J, et al. In-Vivo Experimental Research into the Pre- and Post-Operative Behavior of the Scoliotic Spine. Human Movement Science. 1997 Apr;16(2):299-308.

[9] Michaud F, Lugrís U, Cuadrado J. Determination of the 3D Human Spine Posture from Wearable Inertial Sensors and a Multibody Model of the Spine. Sensors. 2022 Jan;22(13):4796.

[10] Kok M, Hol JD, Schön TB. Using Inertial Sensors for Position and Orientation Estimation. Foundations and Trends® in Signal Processing. 2017 Nov;11(1-2):1-153.

[11] Wang W, Peng Y, Sun Y, Wang J, Li G. Towards Wearable and Portable Spine Motion Analysis Through Dynamic Optimization of Smartphone Videos and IMU Data. IEEE Journal of Biomedical and Health Informatics. 2024 Oct;28(10):5929-40.

[12] Csuhai ÉA, Nagy AC, Váradi Z, Veres-Balajti I. Functional Analysis of the Spine with the Idiag SpinalMouse System among Sedentary Workers Affected by Non-Specific Low Back Pain. International Journal of Environmental Research and Public Health. 2020 Dec;17(24):9259.

[13] Dreischarf B, Koch E, Dreischarf M, Schmidt H, Pumberger M, Becker L. Comparison of Three Validated Systems to Analyse Spinal Shape and Motion. Scientific Reports. 2022 Jun;12(1):10222.

[14] Haddas R, Lawlor M, Moghadam E, Fields A, Wood A. Spine Patient Care with Wearable Medical Technology: State-of-the-Art, Opportunities, and Challenges: A Systematic Review. The Spine Journal. 2023 Jul;23(7):929-44.

[15] Harvey RH, Peper E, Mason L, Joy M. Effect of Posture Feedback Training on Health. Applied Psychophysiology and Biofeedback. 2020 Jun;45(2):59-65.

[16] Posture Pal - Improve & Alert; 2023. https://apps.apple.com/us/app/posture-pal-improve-alert/id1590316152.

[17] Ribeiro DC, Sole G, Abbott JH, Milosavljevic S. Validity and Reliability of the Spineangel® Lumbo-Pelvic Postural Monitor. Ergonomics. 2013 Jun;56(6):977-91.

[18] Mjøsund HL, Boyle E, Kjaer P, Mieritz RM, Skallgård T, Kent P. Clinically Acceptable Agreement between the ViMove Wireless Motion Sensor System and the Vicon Motion Capture System When Measuring Lumbar Region Inclination Motion in the Sagittal and Coronal Planes. BMC Musculoskeletal Disorders. 2017 Dec;18(1):124.

[19] O'Sullivan K, O'Sullivan L, Campbell A, O'Sullivan P, Dankaerts W. Towards Monitoring Lumbo-Pelvic Posture in Real-Life Situations: Concurrent Validity of a Novel Posture Monitor and a Traditional Laboratory-Based Motion Analysis System. Manual Therapy. 2012 Feb;17(1):77-83.

[20] Prajapati S, Sharma JK, Kumar S, Pandey S, Pandey MK. A Review on Comparison of Physical and Mechanical Properties of PLA, ABS, TPU, and PETG Manufactured Engineering Components by Using Fused Deposition Modelling. Materials Today: Proceedings. 2024 May:S2214785324003146.

[21] Riccio C, Civera M, Grimaldo Ruiz O, Pedullà P, Rodriguez Reinoso M, Tommasi G, et al. Effects of Curing on Photosensitive Resins in SLA Additive Manufacturing. Applied Mechanics. 2021 Nov;2(4):942-55.

[22] Vrtovec T, Pernuš F, Likar B. A Review of Methods for Quantitative Evaluation of Spinal Curvature. European Spine Journal. 2009 May;18(5):593-607.

[23] Been E, Kalichman L. Lumbar Lordosis. The Spine Journal. 2014 Jan;14(1):87-97.

[24] Janik TJ, Harrison DD, Cailliet R, Troyanovich SJ, Harrison DE. Can the Sagittal Lumbar Curvature Be Closely Approximated by an Ellipse? Journal of Orthopaedic Research. 1998 Nov;16(6):766-70.

[25] Harrison DE, Janik TJ, Harrison DD, Cailliet R, Harmon SF. Can the Thoracic Kyphosis Be Modeled With a Simple Geometric Shape?: The Results of Circular and Elliptical Modeling in 80 Asymptomatic Patients. Journal of Spinal Disorders & Techniques. 2002 Jun;15(3):213-20.

[26] Panjabi MM, Takata K, Goel V, Federico D, Oxland T, Duranceau J, et al. Thoracic Human Vertebrae. Quantitative Three-Dimensional Anatomy. Spine. 1991 Aug;16(8):888-901.

[27] Intolo P, Carman AB, Milosavljevic S, Abbott JH, Baxter GD. The Spineangel®: Examining the Validity and Reliability of a Novel Clinical Device for Monitoring Trunk Motion. Manual Therapy. 2010 Apr;15(2):160-6.

[28] Seichert N, Baumann M, Senn E, Zuckriegl H. Die Rückenmaus - Ein analog-digitales Meßgerät zur Erfassung der sagittalen Rückenkontur. Physikalische Medizin, Rehabilitationsmedizin, Kurortmedizin. 1994 Apr;04(2):35-43.

[29] Post RB, Leferink VJM. Spinal Mobility: Sagittal Range of Motion Measured with the SpinalMouse, a New Non-Invasive Device. Archives of Orthopaedic and Trauma Surgery. 2004 Apr;124(3):187-92.

[30] Mobbs RJ, Fonseka RD, Natarajan P. Wearable Sensor Technology in Spine Care. Journal of Spine Surgery. 2022 Mar;8(1):84-6.

[31] Hicks GE, George SZ, Nevitt MA, Cauley JA, Vogt MT. Measurement of Lumbar Lordosis: Inter-rater Reliability, Minimum Detectable Change and Longitudinal Variation. Clinical Spine Surgery. 2006 Oct;19(7):501.

[32] Kolber MJ, Pizzini M, Robinson A, Yanez D, Hanney WJ. The Reliability and Concurrent Validity of Measurements Used to Quantify Lumbar Spine Mobility: An Analysis of an iPhone® Application and Gravity Based Inclinometry. International Journal of Sports Physical Therapy. 2013 Apr;8(2):129-37.

[33] ŞahiNoğlu E, ErgiN G, Bakirhan S, Ünver B. Reliability, Concurrent Validity, and Minimal Detectable Change of a Smartphone Application for Measuring Thoracic Kyphosis. Clinical and Experimental Health Sciences. 2023 Sep;13(3):511-6.

[34] Azadinia F, Hosseinabadi M, Ebrahimi I, Mohseni-Bandpei MA, Ghandhari H, Yassin M, et al. Validity and Test–Retest Reliability of Photogrammetry in Adolescents with Hyperkyphosis. Physiotherapy Theory and Practice. 2022 Nov;38(13):3018-26.

[35] Jørgensen R, Ris I, Juhl C, Falla D, Juul-Kristensen B. Responsiveness of Clinical Tests for People with Neck Pain. BMC Musculoskeletal Disorders. 2017 Dec;18(1):548.

[36] Merriaux P, Dupuis Y, Boutteau R, Vasseur P, Savatier X.

A Study of Vicon System Positioning Performance. Sensors.
2017 Jul;17(7):1591.